

LEARNFLOW

Learning Management System (LMS)

Complete UI/UX, Figma Design & Front-End Development Plan

Project Type	Internship Major Project
Frontend	HTML5 • CSS3 • JavaScript (ES6+)
Data & Storage	JSON • LocalStorage
Design Tool	Figma
Target	Responsive Desktop • Tablet • Mobile

Project Vision

LearnFlow is a modern, responsive learning platform where students can discover courses, enroll, watch lessons, complete quizzes, monitor progress, save content, and manage their profiles. The implementation is intentionally front-end focused so that practical HTML, CSS and JavaScript concepts are demonstrated clearly.

1. Project Scope

- User authentication screens: Login and Register.
- Student dashboard with enrolled courses, progress and learning statistics.
- Course discovery with search, categories, difficulty filters and sorting.
- Course details with syllabus, instructor/course information and enrollment.
- Learning area with video/lesson content, chapters, resources and completion tracking.
- Quiz module with question navigation, timer, automatic scoring and result summary.
- Progress dashboard showing course completion, lessons, quizzes and activity.
- Bookmarks/Favorites for saving courses or lessons.
- Notifications for course updates, quiz results and achievements.
- Profile and learning-preference management.
- Responsive layouts for desktop, tablet and mobile.
- LocalStorage persistence for user state, progress, bookmarks, quiz results and preferences.

2. Complete User Flow

Landing / Entry → Login or Register → Dashboard → Explore Courses → Course Details → Start Learning → Lesson / Video → Complete Lesson → Next Lesson or Quiz → Quiz Result → Updated Course Progress → Dashboard

Secondary flows: Dashboard → Bookmarks; Dashboard → Notifications; Dashboard → Profile; Courses → Search / Filter / Sort.

3. Figma File Structure

Figma Page	What to create
01 — Design System	Colors, typography, spacing, buttons, inputs, cards, badges, icons, progress bars
02 — User Flow	Authentication, course, learning and quiz flows
03 — Wireframes	Low-fidelity layouts for all major screens
04 — Desktop UI	High-fidelity desktop screens
05 — Mobile UI	Responsive mobile versions
06 — Prototype	Clickable end-to-end student journey

4. Design System & Color Palette

Token	Hex	Use
Primary	#4F46E5	Main buttons, links, active navigation, key actions
Primary Dark	#3730A3	Headings, hover/active emphasis
Background	#F8FAFC	Application background
Surface	#FFFFFF	Cards, panels, forms
Text	#111827	Primary text
Secondary Text	#6B7280	Descriptions and metadata
Border	#E5E7EB	Card/input/divider borders
Success	#16A34A	Completed lessons, success states
Warning	#F59E0B	Warnings and pending states
Error	#DC2626	Validation and error states

Typography

Font: Inter. H1 32px Bold; H2 24px Semibold; H3 18px Semibold; body 14–16px; small text 12px; buttons 14–16px Semibold. Maintain consistent spacing and component states throughout the application.

5. Core UI Screens

- 01. Login** — Email, password, show/hide password, validation, login action, register link.
- 02. Register** — Name, email, password, confirm password, validation and account creation.
- 03. Dashboard** — Welcome section, continue learning card, statistics, enrolled courses, recommendations.
- 04. Explore Courses** — Search bar, category chips, difficulty/rating/duration filters, sorting and course grid.
- 05. Course Details** — Course image, title, rating, difficulty, duration, description, learning outcomes, syllabus and Start Learning.
- 06. Learning / Video** — Video player, course chapters, current lesson, progress, Overview/Resources/Notes and Previous/Next actions.
- 07. Quiz** — Question number, progress bar, timer, answer options, navigation and submission.
- 08. Quiz Results** — Score, percentage, correct/wrong answers, time taken, review and continue-course actions.
- 09. Progress** — Overall progress, completed courses/lessons/quizzes, course progress bars and weekly activity.
- 10. Bookmarks** — Saved courses/lessons with remove and continue actions.
- 11. Notifications** — Course updates, quiz results, achievements and read/unread states.
- 12. Profile** — Student information, editable preferences and save action.

6. Component Library

- Navigation: Sidebar, topbar, mobile bottom navigation, profile menu.
- Course: Course card, thumbnail, category badge, difficulty badge, rating, bookmark button, progress bar.
- Forms: Text input, password input, select, checkbox/radio, validation message, primary/secondary buttons.
- Learning: Video player area, lesson item, chapter list, progress indicator, resource card, previous/next controls.
- Quiz: Question card, answer option, timer, quiz progress, result card.
- Feedback: Toast, notification item, empty state, loading state and error state.

7. JavaScript Architecture

Keep the application data-driven. Course and quiz data should be represented as JavaScript objects/arrays and rendered dynamically into the DOM. Separate storage logic from page-specific logic so the project remains maintainable.

Suggested structure:

```
data/  
  courses.js  
  quizzes.js  
  users.js  
  
js/  
  app.js  
  auth.js  
  dashboard.js  
  courses.js  
  learning.js  
  quiz.js  
  profile.js  
  storage.js  
  utils.js  
  
css/  
  style.css  
  responsive.css  
  
pages/  
  login.html • register.html • dashboard.html • courses.html  
  course-details.html • learning.html • quiz.html • results.html  
  progress.html • bookmarks.html • profile.html
```

8. LocalStorage Plan

Key	Purpose
currentUser	Currently logged-in student
enrolledCourses	Courses the student has enrolled in
courseProgress	Percentage progress for each course
completedLessons	Completed lesson IDs
bookmarks	Saved course/lesson IDs
quizResults	Score, percentage, time and date for quizzes

Key	Purpose
notifications	Notification items and read/unread state
preferences	Category, difficulty and other UI preferences

Example data model

```
courseProgress = { "javascript-101": 68, "html-css-101": 92, "python-101": 45 }  
  
quizResults = { "javascript-quiz-1": { score: 8, total: 10, percentage: 80, date: "2026-09-07" } }
```

9. Development Flow

01. Requirements finalization
02. User flow and information architecture
03. Low-fidelity wireframes
04. Design system and reusable Figma components
05. High-fidelity desktop screens
06. High-fidelity mobile screens
07. Clickable Figma prototype
08. HTML structure
09. CSS styling and responsive layouts
10. JavaScript data and dynamic rendering
11. Search, filtering and sorting
12. Learning progress and LocalStorage
13. Quiz timer, scoring and results
14. Testing and bug fixing
15. Documentation, screenshots and final presentation

10. Recommended Implementation Priority

Phase 1 — UI Foundation: Design system → navigation → buttons/forms → cards → responsive grid.

Phase 2 — Main Screens: Login → Register → Dashboard → Courses → Course Details → Learning.

Phase 3 — Interactivity: Search → Filter → Sort → Bookmark → Enrollment → Lesson completion.

Phase 4 — Quiz: Questions → answer selection → timer → navigation → scoring → results.

Phase 5 — Persistence: LocalStorage for user state, progress, bookmarks, quiz results and preferences.

Phase 6 — Polish: Notifications → profile → empty/error states → accessibility → responsive testing.

11. Quality & UX Checklist

- Use consistent spacing, typography, colors and component states.
- Maintain clear visual hierarchy and sufficient contrast.
- Provide visible focus/hover/active states for interactive elements.
- Use labels and validation messages for forms.
- Do not rely on color alone to communicate status.
- Make buttons and controls comfortable to use on mobile.
- Ensure course progress is always visible where relevant.
- Handle empty states, invalid input and expired quiz timers.
- Test navigation and LocalStorage persistence after refresh.
- Check layouts at desktop, tablet and mobile breakpoints.

12. Final Project Deliverables

1. Figma design system and high-fidelity screens.
2. Clickable Figma prototype.
3. Responsive HTML/CSS/JavaScript LMS application.
4. LocalStorage-based data persistence.
5. Working search/filter/sort, bookmarks, progress tracking and quiz system.
6. Project documentation and screenshots.
7. Final internship presentation/demo.

Implementation principle

Design first → prototype → build reusable UI → implement data → add interactions → persist state → test → document. Follow this exact sequence so the Figma design and final implementation remain visually and functionally aligned.